

Disagreement-Aware Multi-View Stacking for Robust Malware Classification

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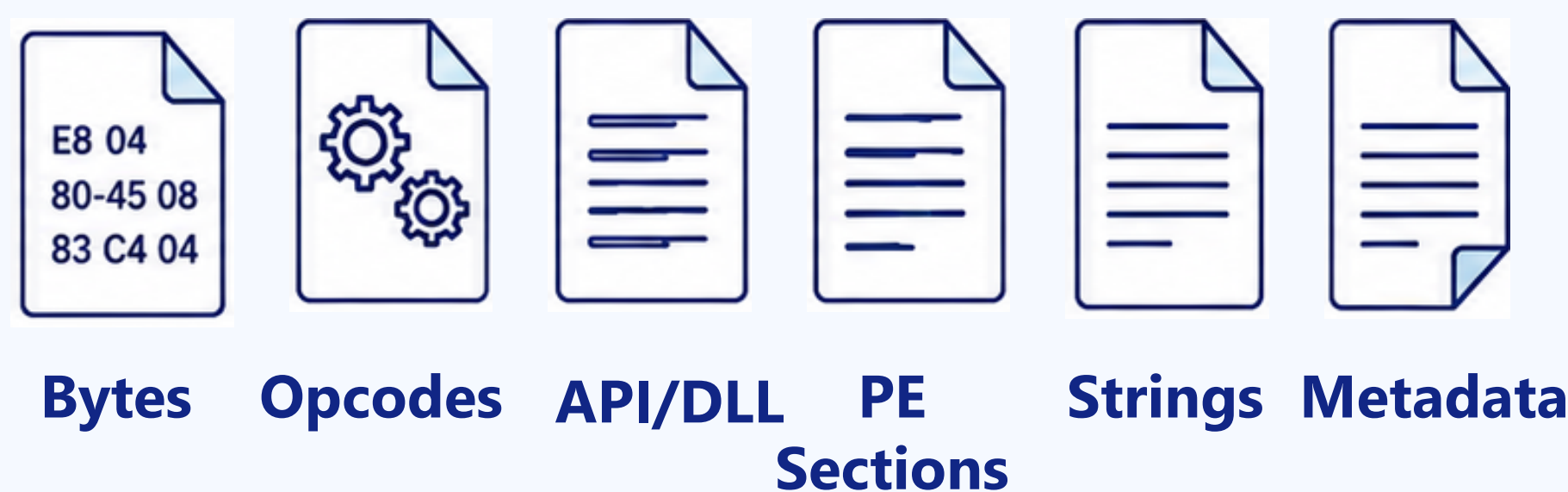
QR code
(Optional)

Scan for code & data

INTRODUCTION

Why Multi-View Malware Classification?

Static malware samples can be represented through raw bytes, opcodes, API/DLL calls, PE sections, printable strings, and structured metadata. Each view captures different characteristics of the same sample.



Why Does Disagreement Matter?

Conventional stacking uses base-model probabilities, but it does not explicitly describe how strongly the models agree or disagree.

High agreement

0.82
0.80
0.78

0.80 mean
Low dispersion

High disagreement

0.99
0.80
0.61

0.80 mean
High dispersion

Takeaway: Same mean, different agreement structure.

Our Contributions

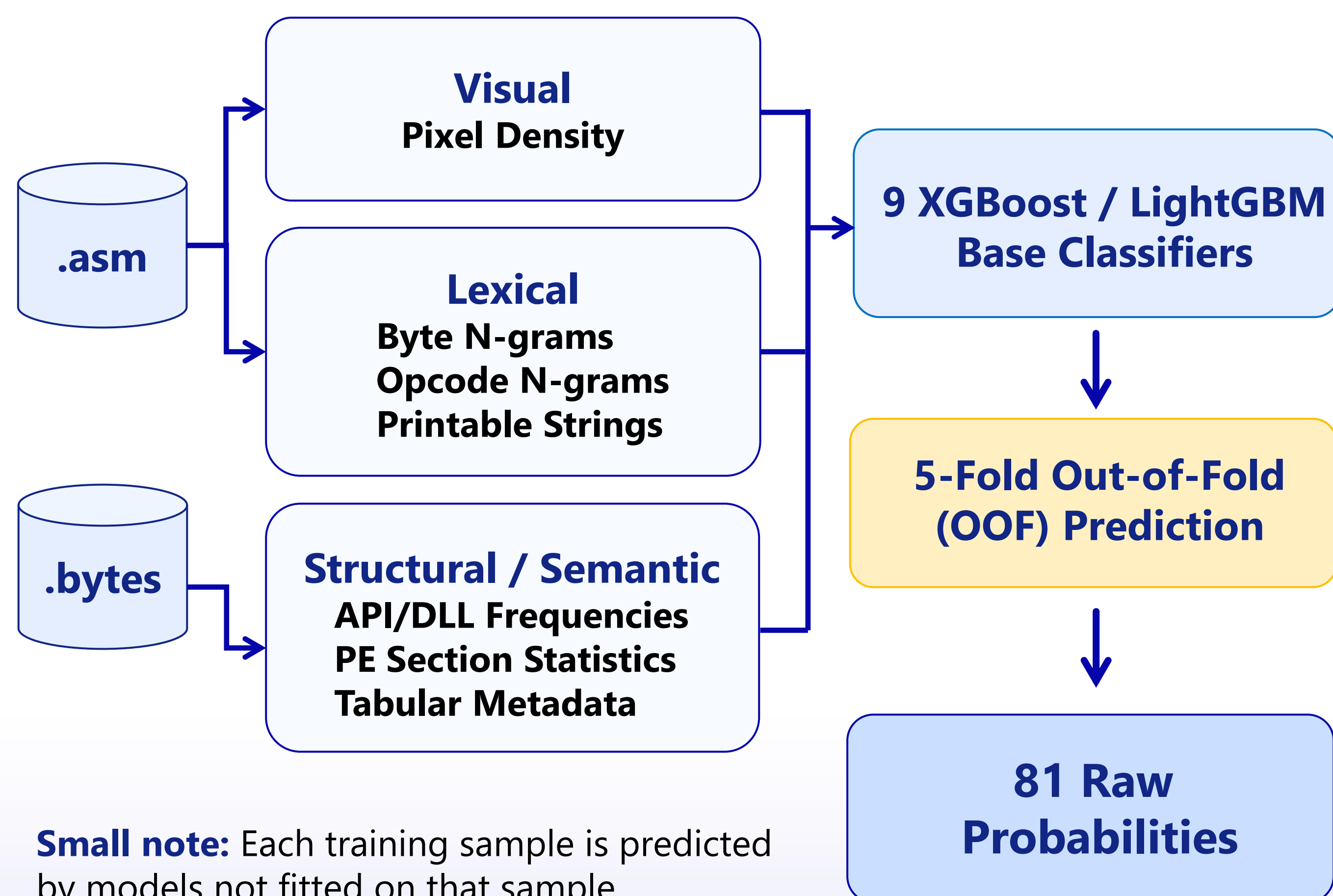
- Multi-view representation:** Seven complementary static feature views from paired .bytes and .asm files.
- Disagreement-aware stacking:** Consensus, dispersion, and uncertainty are explicitly represented at the meta level.
- Optimized fusion:** Two Level-1 classifiers are combined through constrained SLSQP optimization.



Microsoft BIG 2015
10,868 samples 9 malware families

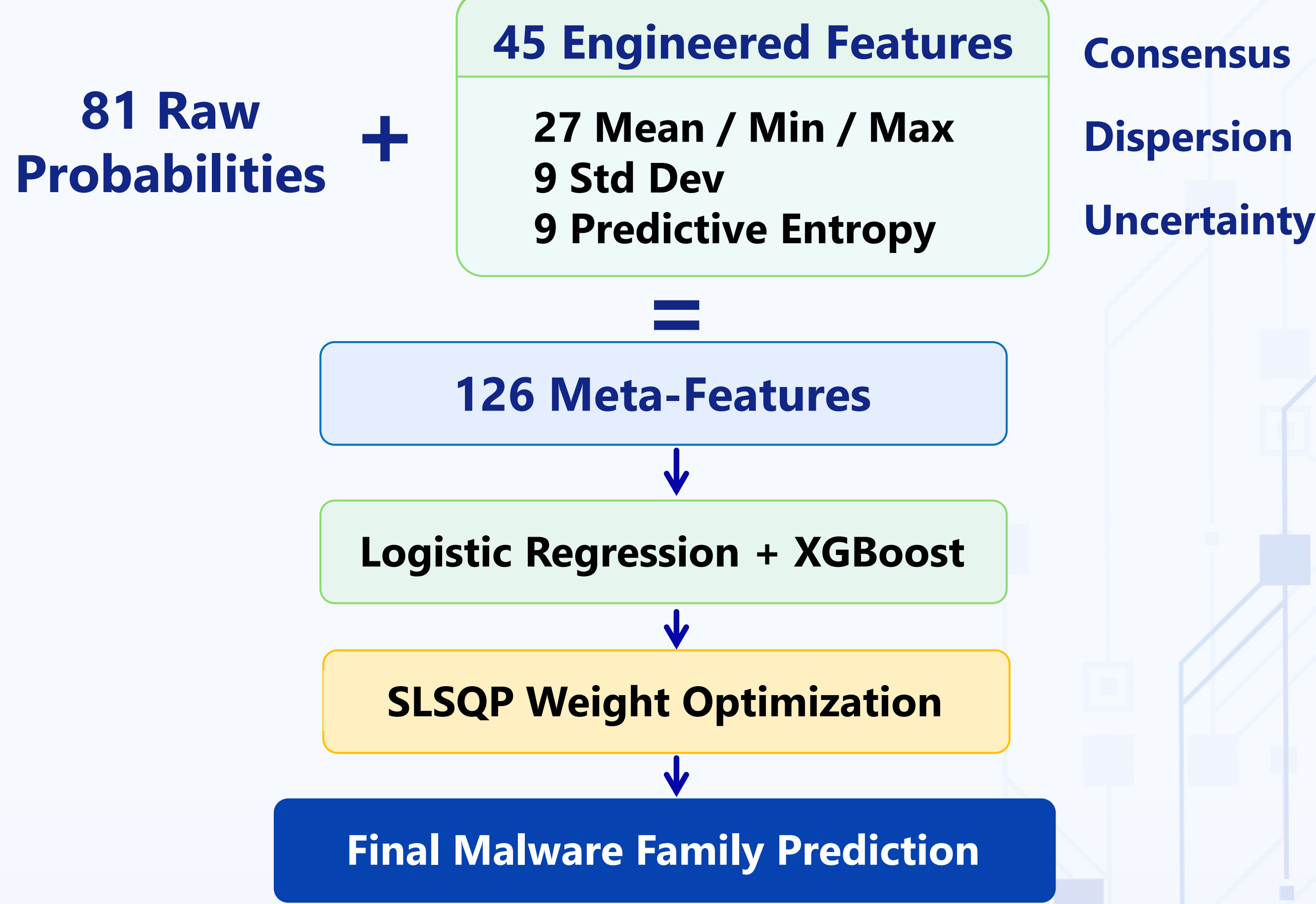
METHOD

STAGE 1 — MULTI-VIEW BASE MODELS



Small note: Each training sample is predicted by models not fitted on that sample.

STAGES 2-3 — DISAGREEMENT-AWARE STACKING



Small note: Non-negative weights that sum to one and minimize multiclass log loss.

RESULT

MAIN METRICS

Performance on the 80/20 Split

99.86%
Accuracy

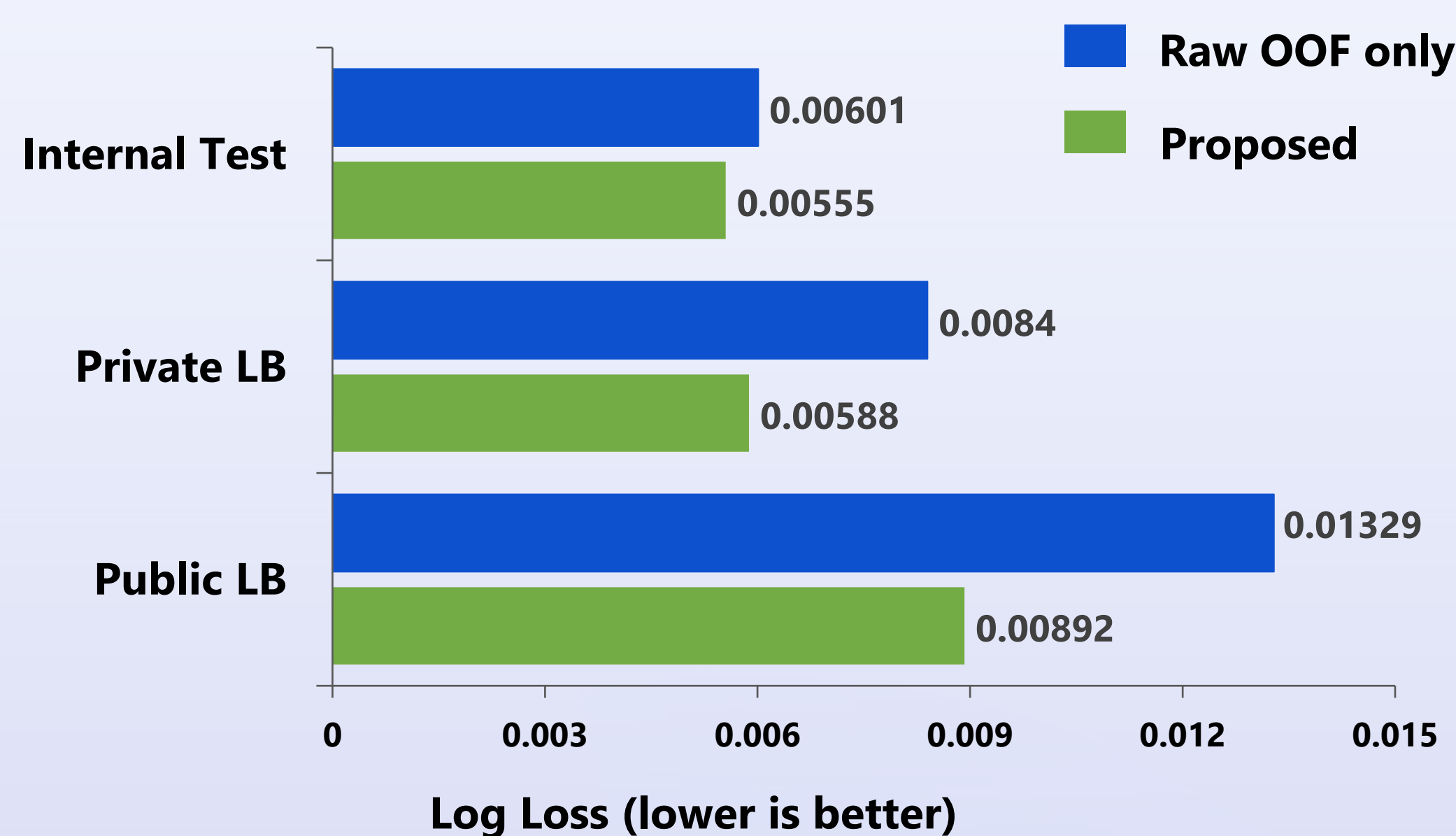
99.83%
Weighted F1-score

0.9984
MCC

0.00555
Test Log Loss

Caption: Lower log loss is better.

KEY ABLATION



Takeaway: Disagreement-aware meta-features reduce log loss across all three evaluations.

SUPPORTING FINDINGS

- Removing the byte N-gram view increases test log loss to 0.01002.
- Removing SLSQP fusion increases test log loss from 0.00555 to 0.00589.
- Full-data leaderboard log loss: 0.00467 (private) 0.00805 (public)

Take-Home Message: Explicitly modeling consensus, dispersion, and uncertainty strengthens multi-view stacking for malware classification on Microsoft BIG 2015.

Scope: Evaluated on Microsoft BIG 2015 using static representations.